Aakash Vasant Bhandary

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EDUCATION

Master of Computer Science *August 2024 - May 2026*
North Carolina State University, Raleigh, NC *GPA - 4.0/4.0*

- **Coursework**: Software Engineering, Algorithms, Generative AI, Neural Networks

Bachelor of Technology - Computer Engineering *August 2020 - May 2024*
Pandit Deendayal Energy University, Gandhinagar, India *GPA - 9.80/10.0 (Dean's List)*

- **Coursework**: Data Structures and Algorithms, Database Management Systems, Computer Networks, Operating Systems, Artificial Intelligence, Machine Learning, Data Analytics with Python

WORK EXPERIENCE

Software Engineer (Volunteer) *September 2026 - Present*
Future World Alliance *Raleigh, NC (Remote)*

- Developed a **Python** and **FastAPI** service that turns paid course enrollments into learner accounts and onboarding emails.
- Engineered a retry-enabled **PostgreSQL** job queue and load tested it with **1,000** enrollments across 3 workers, finishing in 13.7 seconds with zero duplicates.
- Packaged the service with **Docker** and set up **GitHub Actions** CI for linting, tests, and migration checks.
- Collaborated in a cross-functional team in **Agile** sprints to scope three product requirement documents to launch-critical features.

Data Science Intern *May - July 2023 (Part-time)*
AEON Software Pvt. Ltd. *Gandhinagar, India (Remote)*

- Engineered a **Python** pipeline using **PySpark** to process a **4M+** record transportation systems dataset, replacing a pandas-based approach that repeatedly crashed the environment from memory exhaustion.
- Deployed a Python forecasting model with 95.93% R2-score as a **Flask API** on AWS **EC2** using Docker, creating a reusable internal tool to support the company's future needs.
- Created **Power BI** dashboards by writing **SQL** queries after loading the dataset into SQLite, prototyping a workflow for planned live-data integration.

Software Engineer Intern *May - June 2023 (Part-time)*
eInfochips (An Arrow Company) *Gandhinagar, India (Remote)*

- Developed product listing and user cart pages for a client eCommerce site using PHP, **JavaScript**, HTML5/CSS3, and **MySQL**, supporting product browsing and cart management.
- Led weekly client meetings to gather requirements and turn feedback into technical specifications, and coordinate development priorities, contributing to project delivery one week ahead of schedule.
- Configured a **GitHub Actions** CI/CD pipeline on AWS EC2 and RDS to reduce deployment time by **40%**, accelerating client feedback cycles.

PROJECTS

Agentic Chatbot Failure Detection System - GitHub Link

- Designed a rule-based system in **Python** that inspects **tool-calling** traces on a Llama 3.1 8B based chatbot hosted through Ollama.
- Wrote a failure injection harness that corrupts real traces with known failure types and proves each one gets caught by the flagging layer.
- Validated the system against **35+** hand-labelled questions, measuring precision, recall, and F1-score per check.

Financial Transaction Analysis Platform - GitHub Link

- Developed an application in Python, TypeScript, Django REST API framework, and PostgreSQL to add and analyze financial transaction data, isolated per organization, and through role-based access control.
- Integrated a CI/CD pipeline via Jenkins and Docker, running 160+ pytest tests on every commit.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, HTML, CSS, JavaScript (TypeScript)
Full-Stack: React.js, Node.js, Django, Express.js, REST API, Flask, MongoDB, MySQL, PostgreSQL, Agile
Cloud/Deployment: Docker, Jenkins, AWS, Azure, Git/GitHub, Linux
AI/ML: RAG, MCP, PyTorch, TensorFlow, SciPy, Numpy, Pandas, scikit-learn, Claude Code, OpenAI Codex